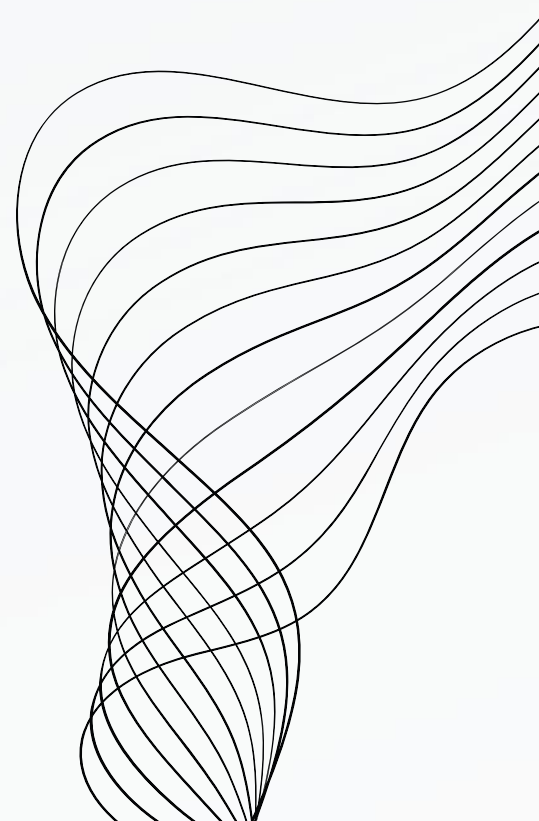


DATAVIS

LECTURE №3. CHANNELS. SYMBOLS. FORMS.





CONTENT



01

SYMBOLS AND CHANNELS

02

TYPES OF CHANNELS AND
SYMBOLS

03

USING FORMS

04

VARIABILITY OF FORMS

05

RELATIVE SIZE

LECTURE QUOTE

Cole Naflic

"The more elements look different, the less likely each one is to attract attention. The idea can be formulated another way: if you want to draw attention to something really important, it should stand out from the rest of the information."

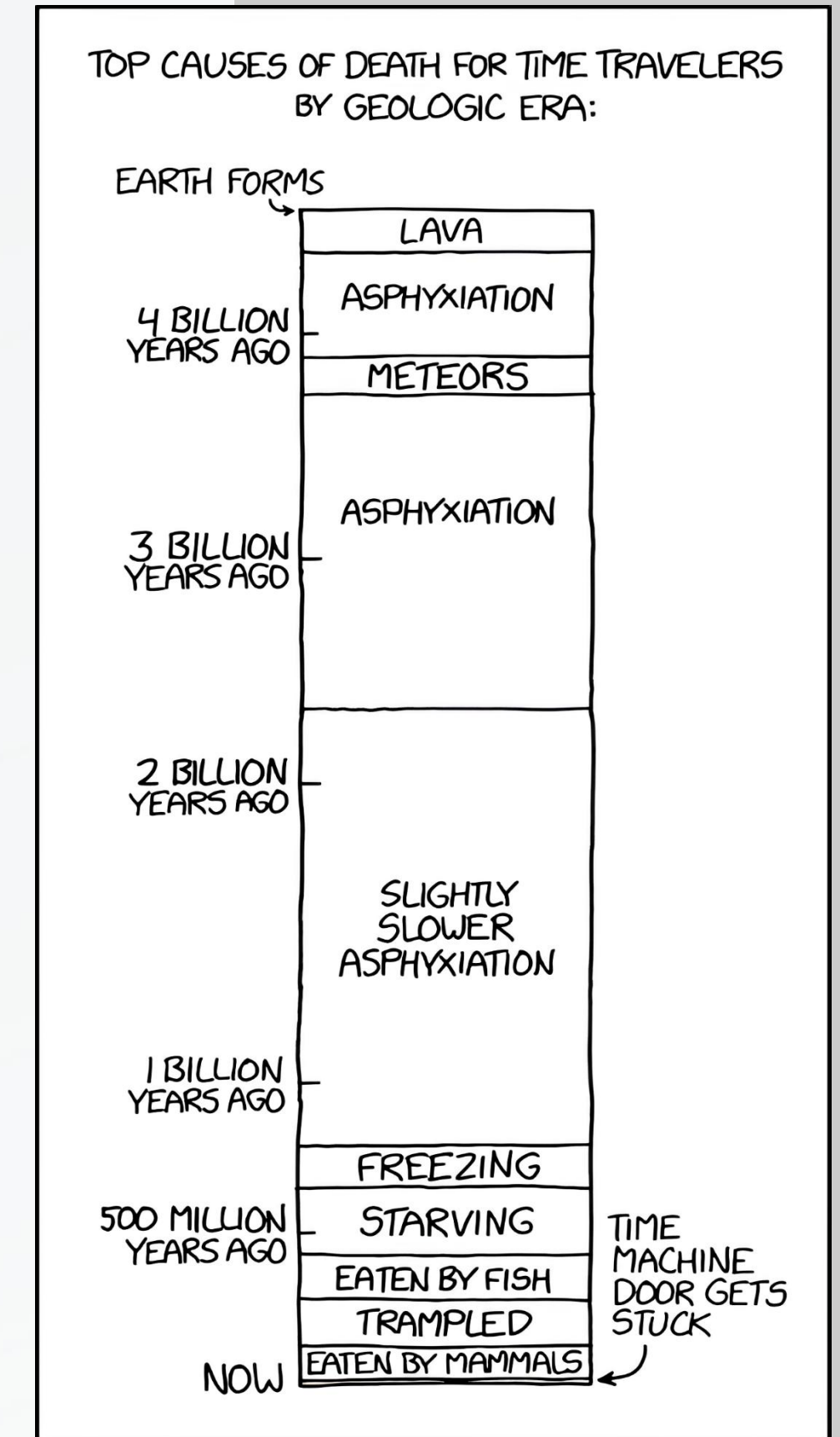


Author of the book "Storytelling with Data: A Data Visualization Guide for Business Professionals" 2018

SYMBOLS AND CHANNELS

SYMBOLS: Represent elements or relationships

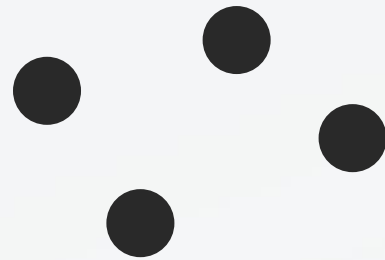
CHANNELS: Change appearance depending on properties



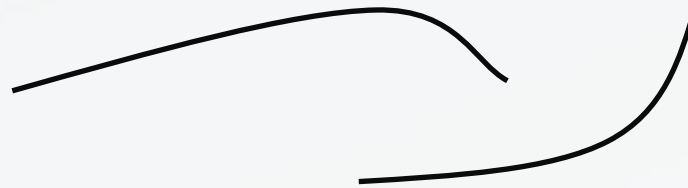
TYPES OF SYMBOLS

SYMBOLS FOR ELEMENTS

Dots



Lines

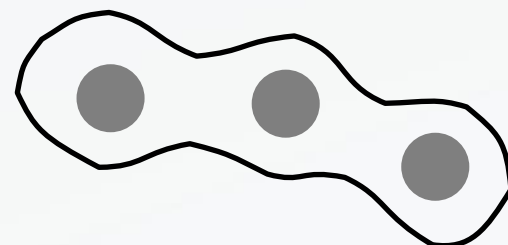


Zones

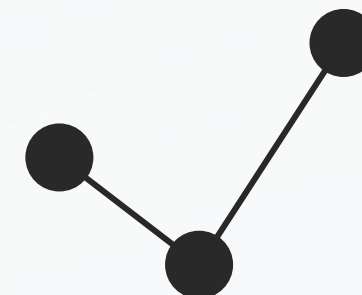


SYMBOLS FOR CONNECTIONS

Zoning



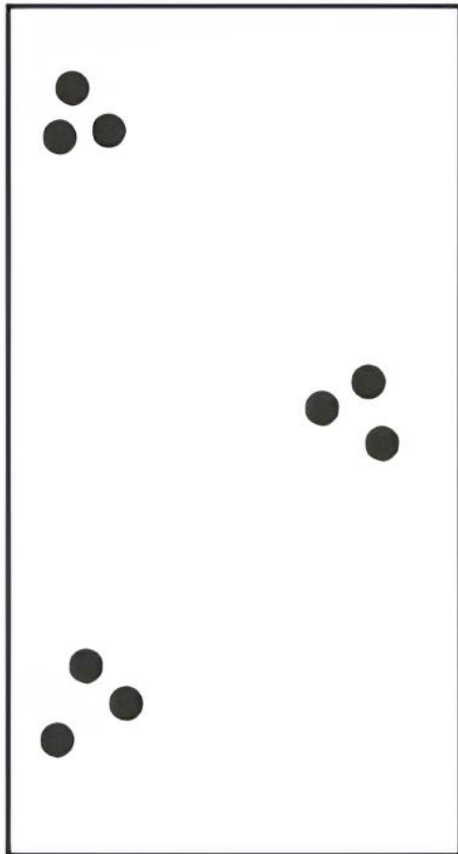
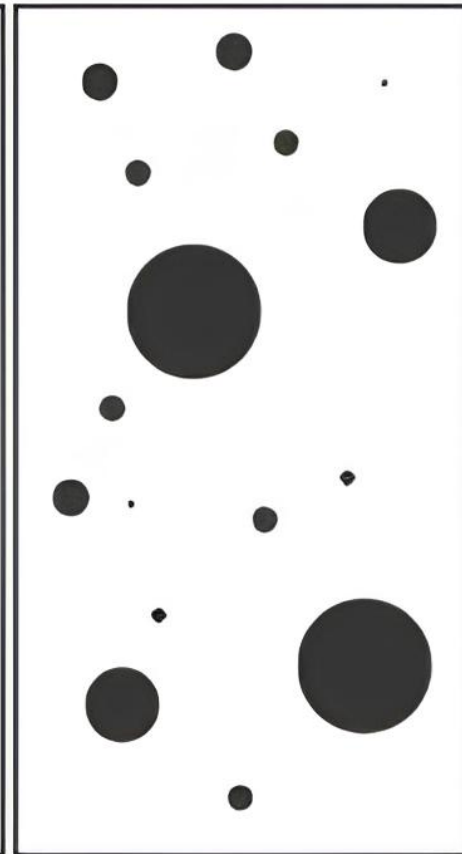
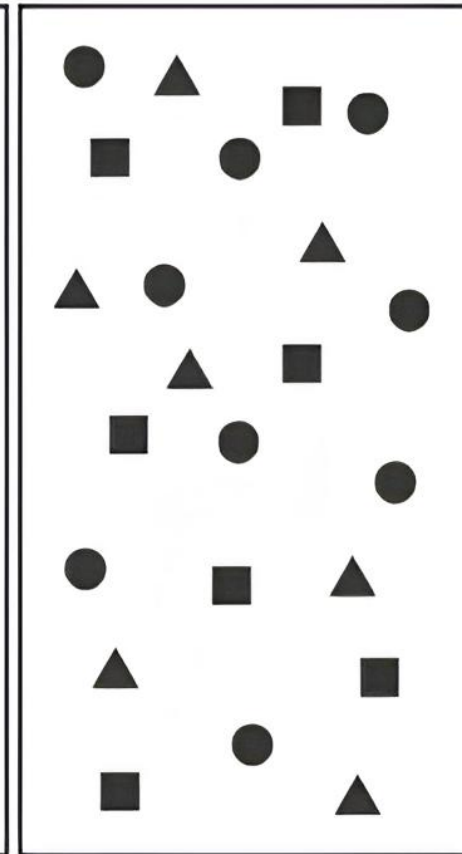
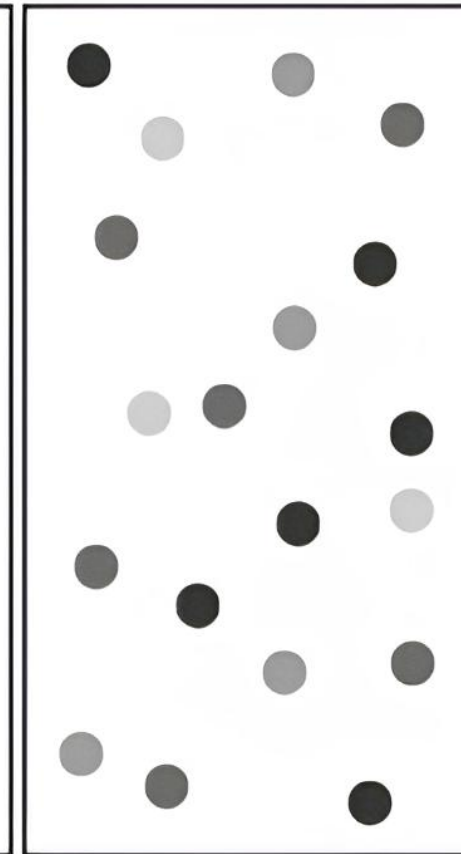
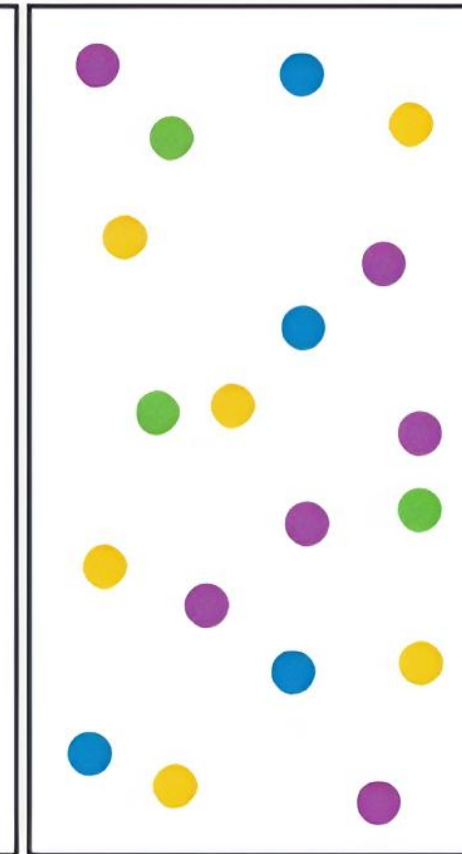
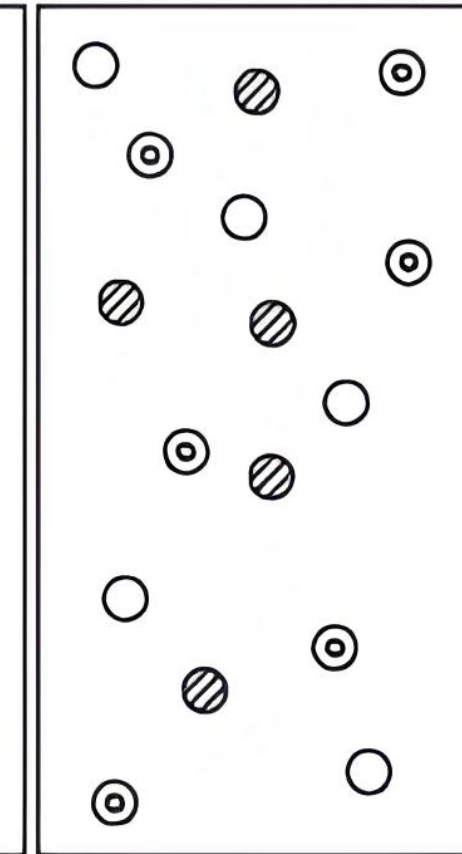
Compound





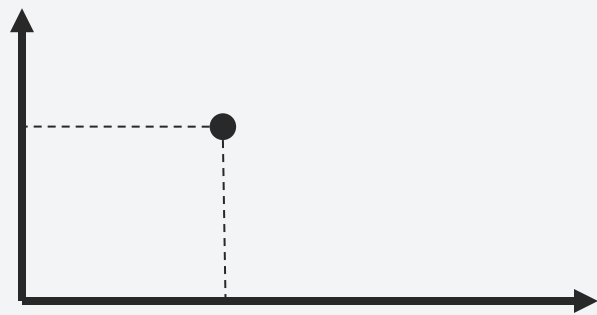
. 6, pp. 1090-1099,
Nov.-Dec. 2010

VISUAL CHANNEL SYSTEM OF JACO BERTIN

POSITION	SIZE	SHAPE	VALUE	HUE	ORIENTATION	TEXTURE
						
Selective Associative Ordered Quantitative	Selective Ordered Quantitative	Associative	Selective Ordered Quantitative	Selective Associative	Selective Associative (sometimes)	Selective Associative Ordered (sometimes)

DIVISION OF CHANNELS BY FEATURES

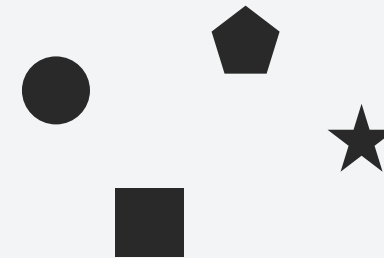
Position



Color



Form

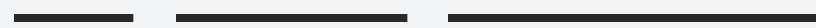


Incline

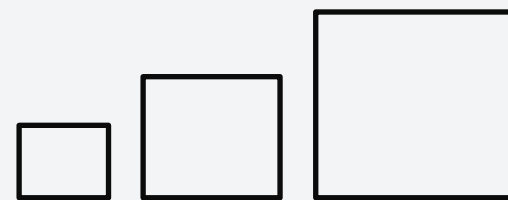


Size

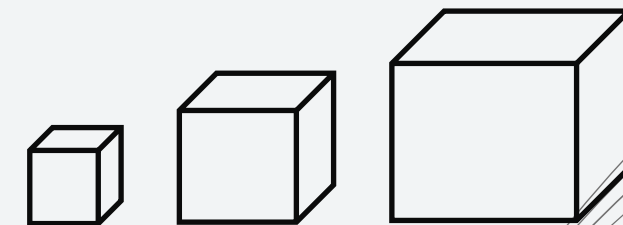
Length



Zone

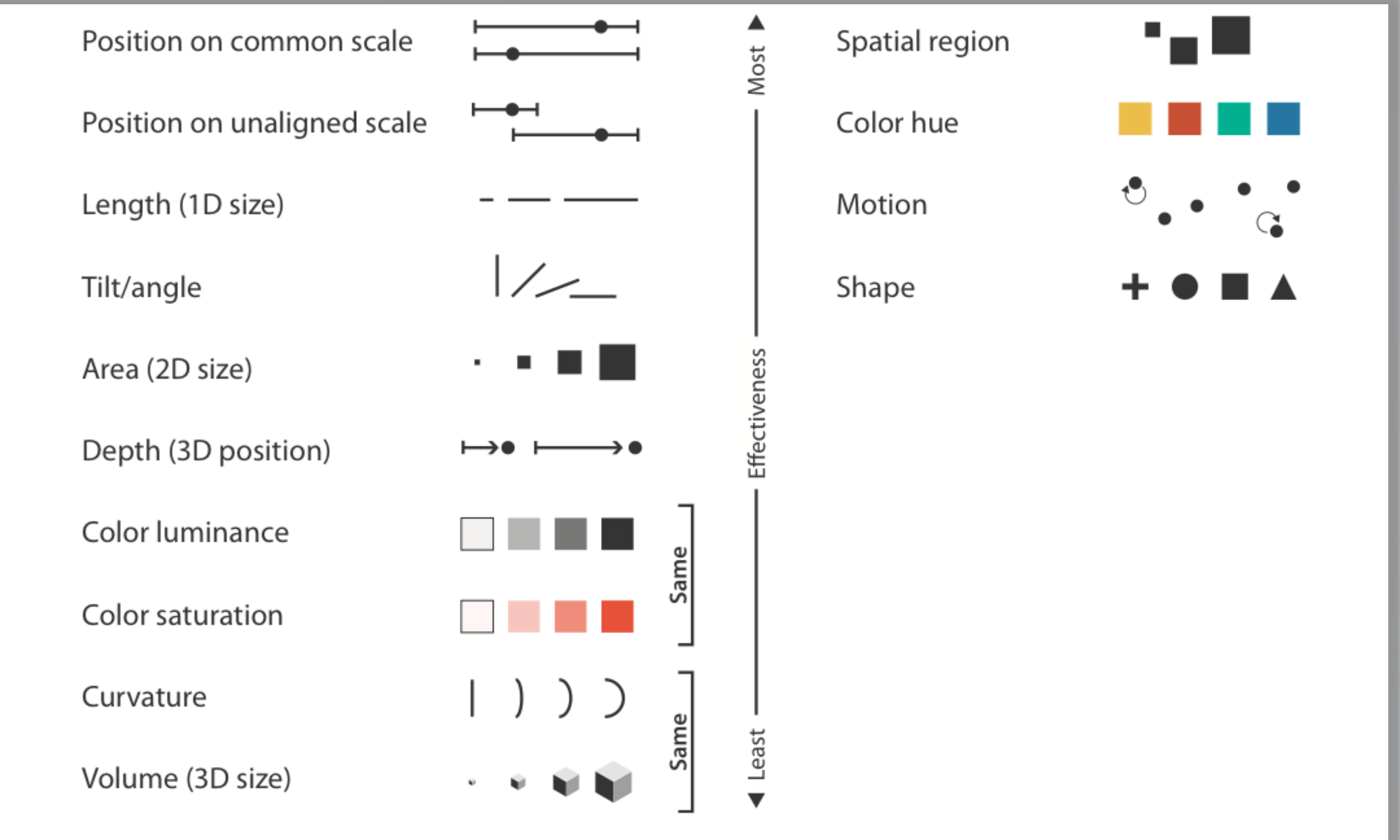


Volume



CHANNELS: TYPES AND EFFICIENCY

O
r
d
e
r
i
b
u
t
e
s



C
a
t
e
g
o
r
i
c
a
l

CHANNEL FEATURES

Size

- Good for 1D;
- Normal for 2D;
- Bad for 3D.

*It is easy to distinguish by size.
The stripes need to be aligned
by position.*



Color

- Suitable for qualitative (categorical) data;
- Suitable for grouping;
- Practically unsuitable for quantitative data.

*Only a limited number (up to 7)
can be used.
Requires caution.*

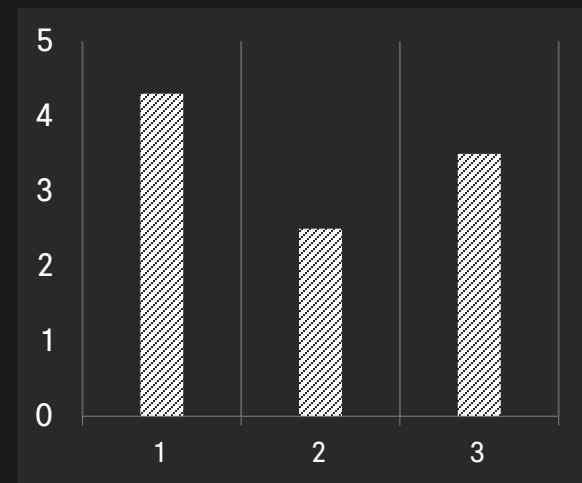


Form

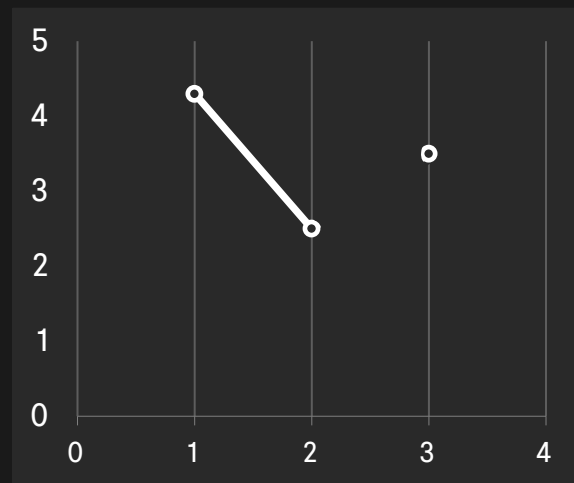
- Suitable for a wide range of classes.
- It is well composed with flowers
- .No grouping and no sorting.



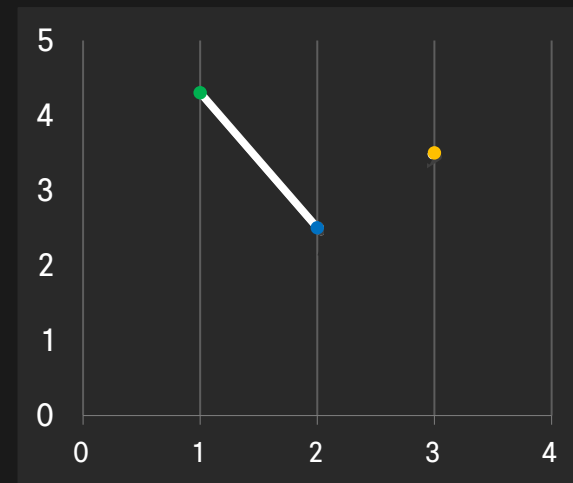
ADDING PROPERTIES



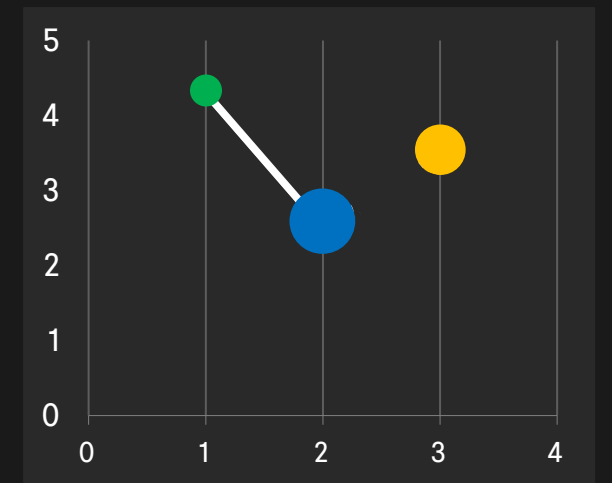
2 PROPERTIES



3 PROPERTIES



4 PROPERTIES

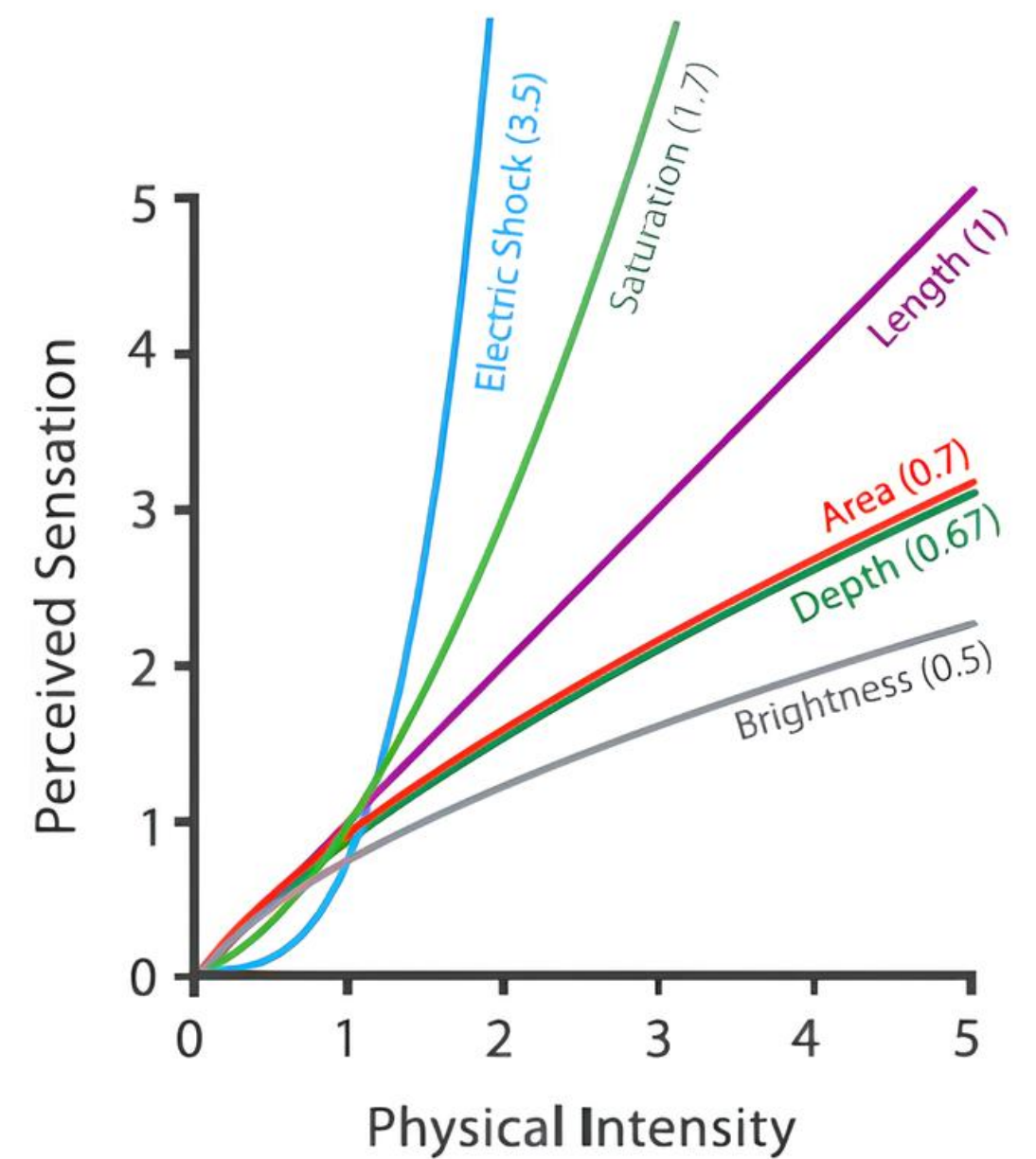


5 PROPERTIES

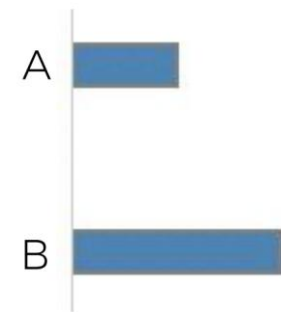
Stevens' Law

$$\psi(I) = kI^a$$

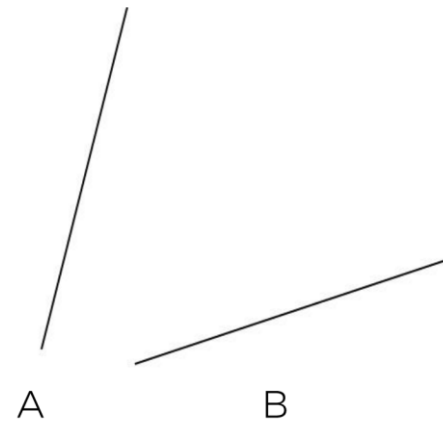
- I (intensity) – stimulus strength in physical quantities;
- a – an exponential function depending on the type of stimulation;
- k – a proportional constant depending on the quantities used;



HOW DIFFERENT IS IT?



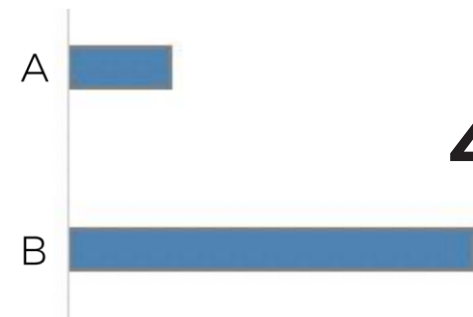
2 times



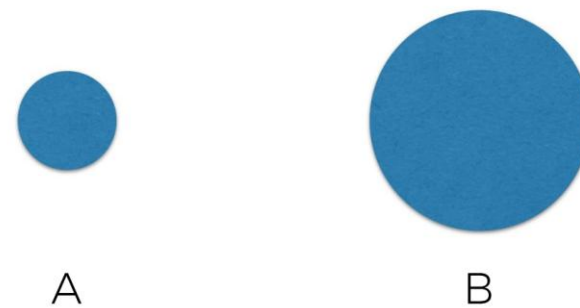
4 times



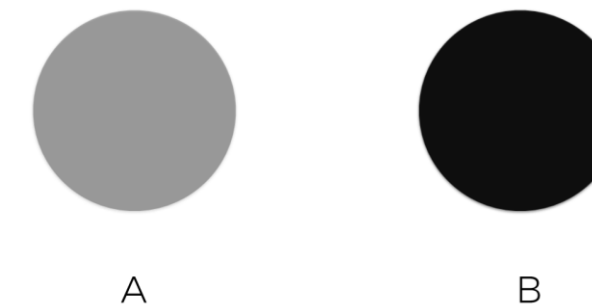
2 times



4 times

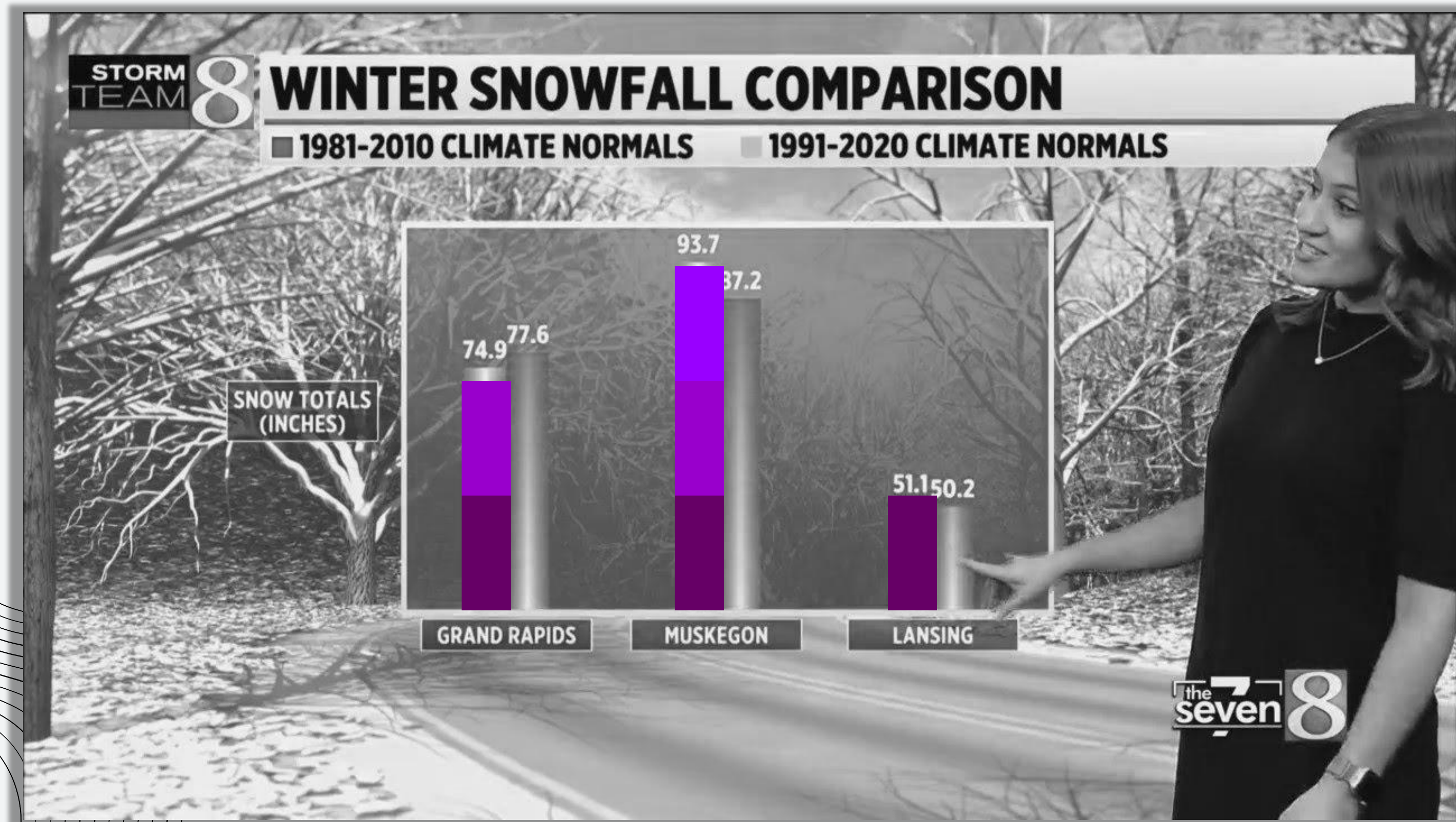


4 times



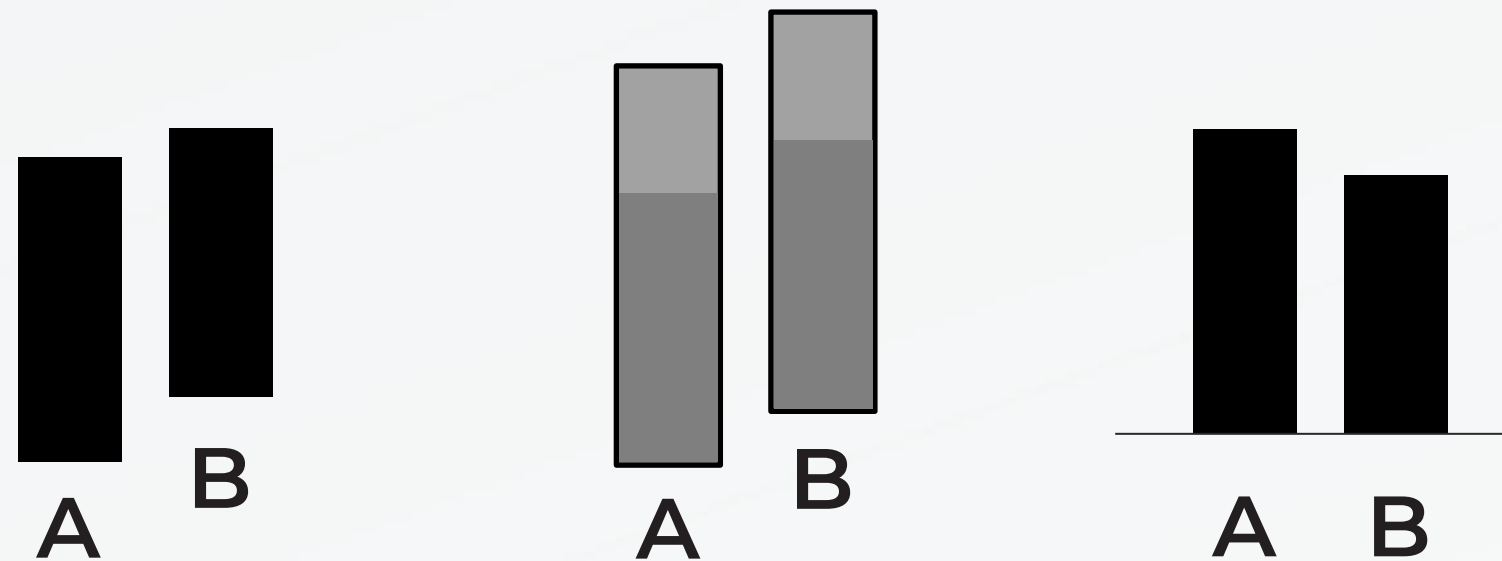
3 times

GOOD HISTOGRAM?



OTHER FACTORS AFFECTING COMPARISON ACCURACY

- Alignment;
- Distance;
- Distractions;
- General scale.



Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods

WILLIAM S. CLEVELAND and ROBERT MCGILL*

The subject of graphical methods for data analysis and for data presentation needs a scientific foundation. In this article we take a few steps in the direction of establishing such a foundation. Our approach is based on *graphical perception*—the visual decoding of information encoded on graphs—and it includes both theory and experimentation to test the theory. The theory deals with a small but important piece of the whole process of graphical perception. The first part is an identification of a set of *elementary perceptual tasks* that are carried out when people extract quantitative information from graphs. The second part is an ordering of the tasks on the basis of how accurately people perform them. Elements of the theory are tested by experimentation in which subjects record their judgments of the quantitative information on graphs. The experiments validate these elements but also suggest that the set of elementary tasks should be expanded. The theory provides a guideline for graph construction: Graphs should employ elementary tasks as high in the ordering as possible. This principle is applied to a variety of graphs, including bar charts, divided bar charts, pie charts, and statistical maps with shading. The conclusion is that radical surgery on these popular graphs is needed, and as replacements we offer alternative graphical forms—*dot charts*, *dot charts with grouping*, and *framed-rectangle charts*.

KEY WORDS: Computer graphics; Psychophysics.

1. INTRODUCTION

Nearly 200 years ago William Playfair (1786) began the serious use of graphs for looking at data. More than 50 years ago a battle raged on the pages of the *Journal of the American Statistical Association* about the relative merits of bar charts and pie charts (Eells 1926; Croxton 1927; Croxton and Stryker 1927; von Huhn 1927). Today graphs are a vital part of statistical data analysis and a vital part of communication in science and technology, business, education, and the mass media.

Still, graph design for data analysis and presentation is

largely unscientific. This is why Cox (1978) argued, "There is a major need for a theory of graphical methods" (p. 5), and why Kruskal (1975) stated "in choosing, constructing, and comparing graphical methods we have little to go on but intuition, rule of thumb, and a kind of master-to-apprentice passing along of information. . . . there is neither theory nor systematic body of experiment as a guide" (p. 28–29).

There is, of course, much good common sense about how to make a graph. There are many treatises on graph construction (e.g., Schmid and Schmid 1979), bad practice has been uncovered (e.g., Tufte 1983), graphic designers certainly have shown us how to make a graph appealing to the eye (e.g., Marcus et al. 1980), statisticians have thought intensely about graphical methods for data analysis (e.g., Tukey 1977; Chambers et al. 1983), and cartographers have devoted great energy to the construction of statistical maps (Berlin 1973; Robinson, Sale, and Morrison 1978). The ANSI manual on time series charts (American National Standards Institute 1979) provides guidelines for making graphs, but the manual admits, "This standard . . . sets forth the best current usage, and offers standards 'by general agreement' rather than 'by scientific test'" (p. iii).

In this article we approach the science of graphs through human graphical perception. Our approach includes both theory and experimentation to test it.

The first part of the theory is a list of elementary perceptual tasks that people perform in extracting quantitative information from graphs. In the second part we hypothesize an ordering of the elementary tasks based on how accurately people perform them. We do not argue that this accuracy of quantitative extraction is the only aspect of a graph for which one might want to develop a theory, but it is an important one.

The theory is testable; we use it to predict the relative performance of competing graphs, and then we run experiments to check the actual performance. The experiments are of two types: In one, once the graphs are drawn, the evidence appears so strong that it is taken *prima facie* to have established the case. When a strong effect is perceived by the authors' eyes and brains, it is likely that it will appear to most other people as well. In

* William S. Cleveland and Robert McGill are statisticians at AT&T Bell Laboratories, Murray Hill, NJ 07974. The authors are indebted to John Chambers, Ram Gnanadesikan, David Krantz, William Kruskal, Colin Mallows, Frederick Mosteller, Henry Pollak, Paul Tukey, and the JASA reviewers for important comments on an earlier version of this article.

Cleveland, W. S., & McGill, R. (1984). Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods. *Journal of the American Statistical Association*, 79(387), 531–554

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